Author Contribution Statement

Extending the SAMBPS Estimation Window Beyond 150 ms: A Nine-State VSM Model with Full AVR and PSS Dynamics

CRediT Taxonomy

Contribution	**A. V. Eluvathingal**	**R. K. Prasad**
Conceptualisation	Lead	Supporting
Formal analysis	Lead	—
Funding acquisition	—	Lead
Investigation	Lead	—
Methodology	Lead	Supporting
Project administration	—	Lead
Software	Lead	—
Supervision	—	Lead
Validation	Lead	Supporting
Visualisation	Lead	—
Writing — original draft	Lead	—
Writing — review & editing	Supporting	Lead

Plain-Language Statement (for portal upload)

Arun V. Eluvathingal: Identified the 150 ms estimation-window limitation of the sixth-order Park dq model; formulated the nine-state VSM augmented ODE with IEEE Type-1 AVR and lead-lag PSS; derived the observability conditions and proved the window-extension proposition; implemented the Levenberg–Marquardt estimator for the augmented state vector; designed and ran all simulation case studies on the 10-bus islanded microgrid; produced all figures and tables; wrote the manuscript draft.

R. K. Prasad: Supervised the research programme; provided guidance on excitation system modelling and microgrid case study design; reviewed and edited the final manuscript; secured funding through the SAMBP project.

Conflict of interest: The authors declare no conflict of interest.

Data and code availability: Simulation scripts (MATLAB/Simulink and Python) and benchmark system data will be made available via the IIT Madras institutional repository upon acceptance.